

Where passives and PPs meet: the case of Udmurt^{4*}

Ekaterina Georgieva & Éva Dékány

ELTE Research Centre for Linguistics

1. Introduction

It is well-known that constructions called “passive” cross-linguistically do not form a homogeneous class. In this paper we argue that some “passives” are not verbal or adjectival but involve a PP predicate instead. The empirical basis of our investigation is the so-called “passive” *-(e)m̄jn* construction in Udmurt (Uralic), as illustrated in (1).

- (1) Perepeć śi-**em̄jn** (anaj-en).
perepech eat-EM̄JN mother-INS
‘The perepech has been eaten (by the mother).’ (F. Gulyás and Speshilova 2014:70)

Diachronic studies (Yemelyanov 1927, Serebrennikov 1963, Bartens 2000) agree that the *-(e)m̄jn* ending was the concatenation of two suffixes: *-(e)m* and *-jn*. Synchronically, *-(e)m̄jn* occurs in two non-finite environments: *i*) past participial relative clauses, which modify NPs and relativize a theme argument¹ (2) and *ii*) gerunds, which have the distribution of NPs (3). Whether (2) and (3) feature the same *-(e)m̄jn* ending or they involve two distinct suffixes is a matter of controversy.

- (2) [Kil’em ar-j̄n pukt-**em̄**] korka džuz.
last year-INE build-EM house burn.PST.3SG
‘The house that was built last year burned down.’ (Dékány and Georgieva 2020:170)
- (3) [[Azbar-e p̄r-**em̄**-z̄j] bere] pumišk-i-z̄j ta todmotem murt-en.
yard-ILL enter-EM-POSS.3PL after meet-PST-3PL this unknown person-INS
‘They met with this stranger after entering the yard.’ (Alatyrev 1970:153)

*This work was supported by the Hungarian National Research, Development and Innovation Office through grants NKFIH PD 147148 to the first author and NKFIH FK 145985 to the second author. We thank the audiences of NELS56, IGG50, BieLeiCoS and SOUL5, Irina Burukina, Marcel den Dikken, Katalin É. Kiss as well as our native speaker consultants Ekaterina Suntsova and Anna Obraztsova.

¹They may occasionally also relativize locative or instrument NPs, see Brykina and Aralova (2012).

As for the suffix *-jn*, it is clearly a case marker. Synchronically, both the inessive and instrumental case suffixes have an allomorph in *-jn* (see (4) and (5), respectively) and it has been debated which of these two cases was involved in the formation of *-(e)m-jn*. Yemelyanov (1927) and Bartens (2000) view the *-jn* of *-(e)m-jn* as the inessive, while Serebrennikov (1963) and A. Kövesi (1965) take it to correspond to the instrumental.

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|-----|---------------------------------------|-----|---|
| (4) | sumka-jn
bag-INE
'in a/the bag' | (5) | sumka-jen, ki-jn, sumka-os-jn
bag-INS hand-INS bag-PL-INS
'with a/the bag, with a/the hand, with a/the bags' ² |
|-----|---------------------------------------|-----|---|

Recent functional-typological studies have argued that over time, the *-(e)m-jn* sequence was reanalyzed into a new, synchronically non-decomposable suffix: *-(e)mjn* is now treated as a member of the participial paradigm ("resultative participle", Asztalos 2010, F. Gulyás and Speshilova 2014, Asztalos and Szabó 2024). At least for transitive-base verbs, (1) is taken to represent an eventive/stative passive construction (which possibly developed under Russian influence, see also Asztalos 2025, Leinonen and Vilkkuna 2000). We argue that in spite of recent claims to the contrary, the *-(e)mjn* form is not a passive that is similar to the eventive or stative passives in English. Instead, we suggest that *V-(e)m-jn* is syntactically a predicative PP, such that the inessive P head takes a nominalized VP as its complement.

This paper is organized as follows. Section 2 motivates treating *V-(e)m-jn* as a predicative PP headed by the inessive case. Section 3 argues that the complement position of the PP is occupied by a silent noun with a participial RC modifier and explores the extent to which the label "passive" is appropriate for *V-(e)m-jn*. Section 4 rounds out the paper.

2. A PP analysis of the so-called "passive"

We argue against the view that the originally bi-morphemic *-(e)mjn* has fused, grammaticalized into a participial suffix and is now used in English-type passives. We suggest that *-(e)m-jn* is still a composite ending that comprises one of the uses of *-(e)m* in (2)/(3) and the inessive *-jn*, and syntactically it maps onto a nominalized verb in a P-complement position.

As our proposal treats *-(e)m-jn* as an inessive PP, it predicts that *-(e)m-jn* should pattern with inessive PPs and not with participles. In contrast, the view that *-(e)m-jn* has been reanalyzed into a resultative participle predicts the opposite. Below, we will show that *V-(e)m-jn* expressions indeed pattern with inessive PPs based on three diagnostics: external distribution, copula drop and negation.

First, let us discuss the external distribution of *-(e)m-jn*. A defining property of participles in Udmurt is that they occur in adnominal position as non-finite RCs, as was shown in (2), repeated below in (6). However, only those that bear an attributivizer or a nominalizer can occur predicatively; *-(e)m* cannot (7). (An instance of attributivized participles is the negated participles; we will discuss them at the end of this section.)

²Unless noted otherwise, the data were collected from two native speakers of Udmurt (age group: 30–40, raised in Northern Udmurtia, Speaker 1 is currently residing in Hungary while Speaker 2 lives in Udmurtia).

- (6) [kjl'em ar-jn pukt-**em**] korka (7) *Korka [kjl'em ar-jn pukt-**em**].
 last year-INE build-EM house house last year-INE build-EM
 'the house that was built last year' Intended: 'The house was built last year.'

In contrast, inessive PPs show the opposite pattern: they can serve as the main predicate of the clause (8) but they have no adnominal use (9).³

- (8) Ulmo [pp sumka-**jn**]. (9) *[pp azbar-**jn**] vedra
 apple bag-INE yard-INE bucket
 'The apple is in the bag.' Intended: 'the bucket in the yard'

Against this background, consider the distribution of *-(e)m-jn*. We have already seen that it can occur predicatively, as in (1) or (10). Crucially, *-(e)m-jn* has no adnominal use (11). In both of these respects, it matches the distribution of inessive PPs rather than participles.

- (10) Korka [tue pukt-**em-jn**]. (11) *[tue pukt-**em-jn**] korka
 house this.year build-EM-INE this.year build-EM-INE house
 'The house was built this year.' Int.: 'the house that was built this year'

Second, copula drop also supports that *-(e)m-jn* is not verbal. Clauses with non-verbal predicates (including locative PP predicates) feature no overt copula in the present tense (12), regardless of the person and number of the subject (Alatyrev 1970, Asztalos 2025). If *-(e)m-jn* forms correspond to PPs, it is predicted that generalized copula drop will also apply to them. As shown by (14), this prediction is confirmed. As participles do not occur in predicative position, the reanalysis approach does not predict (14)–(15) in the first place.

- (12) Ulmo sumka-jn ∅. (13) Ulmo sumka-jn val.
 apple bag-INE apple bag-INE be.PST
 'The apple is in the bag.' 'The apple was in the bag.'
- (14) Sumka daša-m-jn ∅. (15) Sumka daša-m-jn val.
 bag prepare-EM-INE bag prepare-EM-INE be.PST
 'The bag is prepared.' 'The bag was prepared.'

Third, negation is also a useful test in Udmurt, as its encoding differs in PP predicates and participles (Perevoshchikov 1962, Edygarova 2015). Clauses with non-verbal predicates, including PP predicates, are negated with the particle *evël* (16). In contrast, participles are negated with the attributivizer *-tem* or, in the case of *-(e)m*, with *-te* (17).

³We note that there is no general ban on NP-internal PP modifiers in Udmurt. For instance, the instrumental or relative PPs can occur adnominally (Usacheva and Arkhangelskiy 2018). Crucially, however, inessive PPs are not licensed in this position. (9) is originally a Beserman Udmurt example from Usacheva and Arkhangelskiy (2018), but our consultants confirmed the judgments for standard Udmurt as well. The fact that the distribution of *-(e)m-jn* in adnominal and predicative positions parallels that of inessive PPs rather than instrumental PPs is the primary syntactic reason why we take *-(e)m-jn* to contain the inessive case. Semantically, the inessive contributes the meaning that we have a stable state as opposed to a change of state.

- (16) Ulmo sumka-*in* *evəl*. (17) [pottj-m-*te*] kılbur
 apple bag-INE NEG publish-EM-NEG.ATTR poem
 ‘The apple is not in the bag.’ ‘unpublished poem’

Against this background, let us now discuss the negation of *-(e)m-in*. The PP analysis predicts that *-(e)m-in* should be negated with *evəl*: [subject [PP V-*(e)m-in*] *evəl*]. This is borne out (18). However, according to the literature (Perevoshchikov 1962, Edygarova 2015), *-(e)m-in* can also be negated with the negative participle *-mte*, as in (19). The negation pattern in (19) is expected under the reanalysis approach but not under the PP approach.

- (18) Kılbur **pott-em-in** *evəl*. (19) Kılbur **pottj-m-te**.
 poem publish-EM-INE NEG poem publish-EM-NEG.ATTR
 ‘The poem is not published.’ ‘The poem is unpublished.’

We argue that in spite of appearances, (19) is not the negation of *-(e)m-in* but the predicative use of the negated participle in (17). As shown earlier in this section, participles in Udmurt typically do not occur predicatively. The negated participle is an exception to this. Based on this, we suggest that negated participles are, in fact, derived adjectives. In syntactic terms, this means that the negative attributivizer spells out an *a* head, which tops off the participial phrase (20). Consequently, negated participles are suitable as predicates.

- (20) a. [[_{aP} [V-*m*] a=-*te*] N] ‘unV-ed N’ (for (17))
 b. [subject [_{aP} V-*m-te*]] ‘(X is) unV-ed’ (for (19))

The structure in (20) predicts that the negative attributivizer will always scope over the participial suffix.⁴ From this it follows that the predicatively used negated participle (19) is functionally equivalent to (18), which might have led the earlier studies on Udmurt to falsely assume that *-(e)m-in* can be negated with the negative participle.

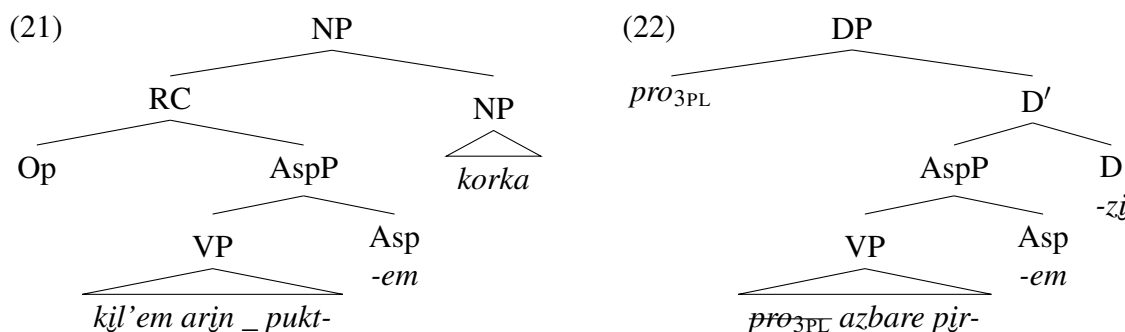
A further and perhaps even stronger argument in favour of the idea that (19) is not the genuine negation of *-(e)m-in* is that unlike *-(e)m-in*, the negated participle does not feature the inessive case. We thus conclude that the negation with *evəl* in (18) is the only true negation of *-(e)m-in*. Therefore, negation also speaks in favour of the PP approach advocated here. In sum, the three diagnostics presented firmly support the idea that *-(e)m-in* corresponds to a PP headed by the inessive case, hence, it is still bi-morphemic.

3. The complement of the inessive P: nominalizing the VP

A comprehensive account of *-(e)m-in* PPs needs to answer two questions: *i*) which *-(e)m* syntax does *-(e)m-in* show, that of participles (2) or gerunds (3); and relatedly, *ii*) what is the internal structure of the nominalized VP that is the complement of the inessive *-in*?

⁴Consequently, the reversative meaning of negation will not be attested, and the participles of verbs like *untie*, *unwrap*, *unwind*, *unlock*, etc., will have the meaning of ‘X has not been V-ed’, i.e., no event occurred, but will lack the meaning of ‘X got unV-ed’. This is confirmed, since unlike English *un-*, the negative suffix *-te* cannot attach to verbs (Perevoshchikov 1962).

Starting with the status of *-(e)m*, we showed in the introduction that this non-finite suffix occurs in two environments: in participial pRCs and in gerunds. In line with Georgieva and Ótrott-Kovács (2016), Georgieva (2018), and Dékány and Georgieva (2020) we provide a unified analysis of the two uses.⁵ The *-(e)m* suffix expones a verbal functional head, which we identify as (anterior) Asp⁰. The reason why both pRCs (2) and gerunds (3) make use of *-(e)m* is that they both contain AspP in their structure. In pRCs we find a truncated (non-finite) VP, which is not topped off by an adjectival or nominal shell. Like finite RCs, pRCs also have a left-peripheral operator (Bruening 2014), which is linked to the gapped theme argument of the verb. The structure of (2) is given in (21). In gerunds, in contrast, the AspP of *-(e)m* is embedded under a D head. In other words, gerunds are mixed verbal/nominal extended projections. D⁰ brings in the nominal properties and hosts the possessive agreement characteristic of gerunds. The verbal structure under D has a fully saturated argument structure (i.e., no argument is gapped). Like non-derived nominals, the gerundive DP bears a structural or semantic case. The structure of (3) is shown in (22).



Among other things, this proposal derives the fact that outwardly gerunds, unlike participles, have a nominal distribution, i.e., appear in argument positions and as complements of Ps. For example, the gerund in (3) is the complement of the P *bere* ‘after’.

We suggest that the same *-(e)m* we find in pRCs and gerunds occurs in *-(e)m-in*, too. Since Ps take nominal complements, the AspP of *-em* in (1) must be nominalized before it can combine with the P of *-in*. The question that now arises concerns the exact mechanism of this nominalization. We have seen that Udmurt can embed an AspP in a phrase with an external nominal distribution in two distinct ways: via a gerund-like syntax or a pRC-like syntax. Since in *-em-in* phrases *-(e)m* spells out Asp and *-in* expones P, we conclude that the nominalization operation itself is not associated with any overt morphology. Thus, under both scenarios the nominalizing element will be null.

We argue that there are empirical arguments supporting the idea that *-em-in* phrases differ from gerunds but bear resemblance to pRCs. These involve the encoding of arguments as well as object idioms. Starting with argument encoding, gerunds obligatorily have an external argument (in the form of an overt genitive NP or a *pro*), which elicits possessive agreement on the non-finite verb (cf. 3PL *-zi* in (3)). In pRCs, on the other hand, the external argument can be omitted (6) and when present, it is encoded as an instrumental PP

⁵Perevoshchikov (1962), Kalinina (2001), Brykina and Aralova (2012), Serdobolskaya et al. (2012) argue that there are two homonymous *-(e)m* suffixes. For arguments against this, see Dékány and Georgieva (2020).

rather than as a possessor (Dékány and Georgieva 2020). However, neither the verb of the pRC, nor the external N head agrees with it. Importantly, *-(e)m-in* phrases typically lack an external argument (10). When it is present, the external argument bears instrumental case and not the genitive case; furthermore, the *V-(e)m-in* complex does not exhibit any agreement (1). In all of these respects, *-(e)m-in* patterns with pRCs rather than gerunds. Gerunds and *V-(e)m-in* also differ in the encoding of the theme argument of a transitive base verb: in gerunds, this argument has accusative case (Dékány and Georgieva 2020), while in the case of *V-(e)m-in* it bears nominative (cf. *perepeć* in (1)).

Moving to object idioms, the data confirm that *-(e)m-in* is fashioned after pRCs. We used the object idiom *vales mertanĭ* ‘to lie in bed/be lazy (lit. measure [the] bed)’. We found that the idiomatic reading is retained in gerunds (23). In contrast, pRCs and *-(e)m-in* phrases only have a non-idiomatic reading, as shown in (24) and (25), respectively.

- (23) [Pet’a-len vales merta-m-ez] anaj-ez-lĭ
 Petya-GEN bed.ACC measure-EM-POSS.3SG mother-POSS.3SG-DAT
 jaraz.
 appeal.PST.3SG
 ‘Petya’s measuring the bed appealed to his mother.’ (non-idiomatic) *or*
 ‘Petya’s lying in bed/being lazy appealed to his mother.’ (idiomatic)

- (24) [merta-m] vales (25) Vales merta-m-in.
 measure-EM bed bed measure-EM-INE
 ‘the measured bed’ ‘The bed is measured.’

Based on these facts, we conclude that *-(e)m-in* phrases do not pattern with gerunds. This means that their structure cannot be a mixed extended projection parallel to (22), with a DP embedded under an inessive P, as in (26):

- (26) [PP [DP [AspP ... verb-(e)m] D=∅] P=*in*] Rejected: *-(e)m-in* as a gerund

Instead, given the similarities between pRCs and *-(e)m-in*, we suggest that they involve a similar structure. Recall that pRCs are extended VPs. However, as nominal modifiers, they occur within an NP, which means that they are contained in a phrase that does have a nominal distribution (21). We thus propose that phrases with *-(e)m-in* map onto such structures, with the NP being headed by a null lexical nominal. This is depicted in (27).

- (27) [PP [NP [RC ... verb-(e)m] [NP N=∅]] P=*in*] (*e)m-in* as an RC

Participial RCs with a null head noun are independently available for certain result nominals formed with *-(e)m* (Dékány and Georgieva 2020). For example, *pīžem* ‘pastry (lit. baked)’ is argued to have the following structure: [[*baked*] N], where N may also be overt: *pīžem make* ‘pastry (lit. baked (some)thing)’.

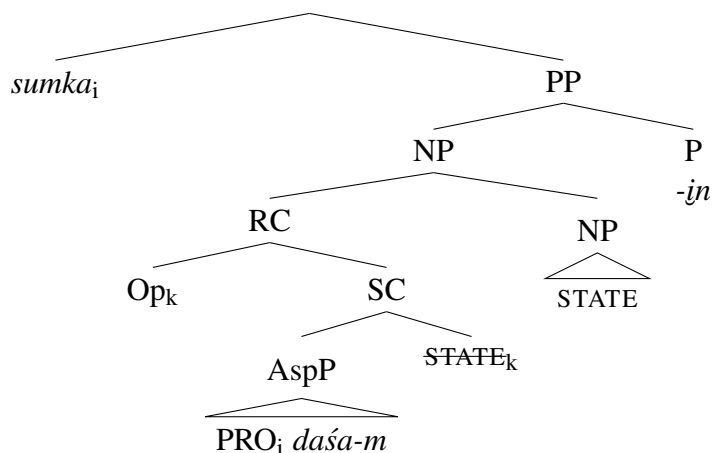
What kind of null lexical N head is involved in (27)? At least for the stative passive reading of *-(e)m-in*, we suggest that it corresponds to a Kaynean silent noun STATE, whose

silence is not a result of ellipsis but it is associated to a null phonology already in the lexicon. The overall meaning of *-(e)m-in* phrases arises compositionally: (1), for instance, is literally ‘The perepech is in an/the eaten state’, and (14) is ‘The bag is in the/a prepared state’.

We argue that apart from (c)overtness of STATE (and the lack of articles in Udmurt), the structure of the Udmurt example in (14), for instance, and that of its English paraphrase, are fully parallel. In both of them, the locative-marked noun ‘state’ is modified by a prenominal past/passive participle *-(e)m* in Udmurt, *-ed* in English). In both languages, the participle employs the gap strategy. This gap is in the internal argument position of the base verb, and the nominal head of the pRC is coferential with this gap (cf. English *the [newly _i built] house_i* or Udmurt (6)).

Udmurt *-(e)m-in* clauses and their English literal translations, ‘X is in a V-ed state’, have a similar structure. We suggest that the pRC modifying STATE embeds a small clause. In the case of (14), the SC corresponds to ‘prepared (is) its state’. That is, the subject of the SC is the AspP of *-(e)m*. The predicate of the SC is the nominal STATE. STATE is being relativized: it is raised to the highest specifier in the relative clause and is linked to the pRC’s external silent head STATE. Thus we are dealing with *predicate relativization*. This is schematized in (28).

(28) *-(e)m-in* as an RC (final version)



Let us now turn to the issue of how it is possible for the NP which occupies the surface subject position of the main clause (in (14): *sumka* ‘bag’) to be also interpreted as the past/passive participle’s internal argument, both in Udmurt and in English ‘The bag is in the/a prepared state’. We suggest that the surface subject is base-generated in the main clause, outside of the PP. From here, it controls a PRO that occupies theme argument position of the the pRC-internal AspP. We indicate this in (28) via coindexation.

Our analysis in (28) partially agrees with Bartens (2000), who suggests that *-(e)m-in* diachronically contains the attributively used participial *-(e)m*. We complemented Bartens (2000) by adducing empirical arguments for this view from argument encoding and object idioms and we put forth a specific formal syntactic proposal as to how exactly pRCs are

integrated into the structure of *-(e)m-in*. In contrast to Bartens (2000), however, we suggest that (28) also represents the synchronic syntax of *-(e)m-in*.

Finally, let us address the question of to what extent, if at all, the label “passive” is appropriate for *-(e)m-in*. The hallmark of passive constructions is that the internal argument is promoted, crossing over the position of the external argument (if the latter is projected at all). In an English-type passive, this movement takes place along the main clausal spine, from the V-complement position to Spec,TP (29). Udmurt *-(e)m-in* clauses have a different structure: here a possibly dropped copula is accompanied by a PP predicate (30).⁶

(29) [TP NP_i [*be* V-*ed* _{-i}]] (30) [TP NP [PP ... V-*(e)m-in*] (*be*)]

While (29) and (30) significantly differ in their structure, there is a sense in which the term “passive” applies to the latter: within the RC, (30) contains an AspP that is itself passive. Whether used in the theme argument relativization (21) or the predicate relativization structure (28), the AspP of *-(e)m* features *i*) a null element in the theme position, which is promoted and becomes a derived subject and *ii*) with transitive base verbs, suppression of the external argument, which either remains implicit or it is included in the form of an adjunct PP.⁷ Expressions with *-(e)m-in* thus have a passive ingredient, but they do not represent the canonical passive structure known from English.

4. Conclusions and broader questions

In this paper we examined the Udmurt “passive” construction *-(e)m-in* and suggested that it is a predicative PP with a nominalized VP as its complement. We further proposed an analysis in which the overall meaning that the surface-subject is *in a target/result state* arises compositionally. The *V-(e)m-in*-construction shows functional similarities to stative passives, which are still the subject of debate, centered around the questions of what the internal structure of participles is and how their arguments are licensed (see Kratzer 2000, Anagnostopoulou 2003, Paslawska and von Stechow 2003, Embick 2004, Bruening 2014, among many others). Our proposal sheds light on the cross-linguistic picture of constructions called “passives”: the *-(e)m-in*-construction may be functionally equivalent to the English or Russian passives, but it has a different syntax.

Going beyond passives, other types of states, e.g., progressives, are also encoded as locative PPs (see Cinque 2017). For instance, the combination of a participial verb and a locative case can be observed in the sparingly used Hungarian progressive. It features the active participle *-ó/ő* and the inessive case (31b).

⁶We are assuming head-final structures for Udmurt, in accordance with its general profile, but this is immaterial to the main point regarding the differences between the *-(e)m-in* construction and English passives.

⁷One of the characteristics of eventive passives is the possibility of expressing the external argument in a PP. As seen in (1), this is possible with *-(e)m-in* as well, though it should be pointed out that compared to the PP-less version, some speakers consider the inclusion of the instrumental PP artificial (see also Asztalos 2025). The instrumental PP can follow the *V-(e)m-in* complex (1). What this means for our analysis is that in the surface structure, this PP is located outside of the predicative PP headed by *-in*. Whether their position is the result of extraposition or of base-generation into the main clause is an issue we leave for further research.

- (31) a. az [emelked-**ő**] árak b. Az árak emelked-**ő-ben** vannak.
 the rise-PTCP prices the prices rise-PTCP-INE be.PRS.3PL
 ‘the rising prices’ ‘Prices are on the rise.’ (personal knowledge)

The Finnish so-called third infinitive series is analogous: it combines the non-finite suffix *-mA* with the inessive case to yield a progressive meaning (Vainikka 1989, Ylikoski 2003). In the French ‘*gérondif présent*’ (whose use is limited to free adjuncts), too, a present participle combines with the locative *P en* ‘in’. Similarly to Udmurt, in these constructions, nominalizing the verb phrase does not have an overt morphological reflex. Our proposal opens the possibility that these may involve attributive modification of a null noun as well.

Our analysis pairs a stative reading with a structure in which a P embeds a nominalized verb. For progressive states, this has well-known analogues, including the Middle English locative-progressive (*had been on hunt-ynge*, Malory: *Le Morte d’Arthur*) and its dialectally still used reduced form (*ahunting*), the Dutch *aan het*-progressive, as well as the colloquial German *am*-progressive and the Austrian German *beim*-progressive. These cross-linguistic parallels provide further support for the analysis advocated here.

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